Questions

Q1.

An electric kettle is used to boil some water.

While the water is boiling, 566 000 J of thermal energy turns 0.250 kg of water into steam.

Calculate the specific latent heat of vaporisation of water.

Use an equation selected from the list of equations at the end of this paper.

(3)

specific latent heat = ... J/kg

(Total for question = 3 marks)

Q2.

Figure 8 shows a small container of carbon dioxide at high pressure.

The pressure, P_1 , in the container is 8.00 MPa.

The volume, V_1 , of the container is 14.5 cm^3.

Figure 8

The container is pierced and all of the carbon dioxide goes into a large balloon.

The volume of gas, V_2, in the large balloon is 1160 cm^3.

Calculate the pressure, P_2, in the large balloon.

Use the equation

$$P_1V_1 = P_2V_2$$

(3)

pressure in the large balloon = ... MPa

(Total for question = 3 marks)

Q3.

A fixed mass of helium gas is compressed into a container, with no change in temperature.

The table in Figure 7 shows the pressure P_1 and volume V_1 before the gas is compressed, and the volume V_2 after the gas is compressed.

	pressure in kPa	volume in m³
before the gas is compressed	$P_1 = 105$	$V_1 = 2.3$
after the gas is compressed	$P_2 =$	$V_2 = 0.20$

Figure 7

(i) Calculate the pressure P_2 after the gas is compressed.

(2)

Use the equation

$$P_2 = \frac{P_1 \times V_1}{V_2}$$

pressure P^2> = ... kPa

(ii) A shop sells the container of helium gas shown in Figure 8.

The container of helium gas can be used to fill party balloons.

Figure 8

A filled party balloon has a volume of 0.07 m^3 at a pressure of 105 kPa.

The shop claims that you can fill at least 30 party balloons using the gas from the container.

Comment on the shop's claim.

Use information about the volume of the gas at a pressure of 105 kPa as shown in the table in Figure 7.

(3)

.....

.....

.....

.....

(Total for question = 5 marks)

Q4.

Figure 16 shows a metal container with a movable piston.

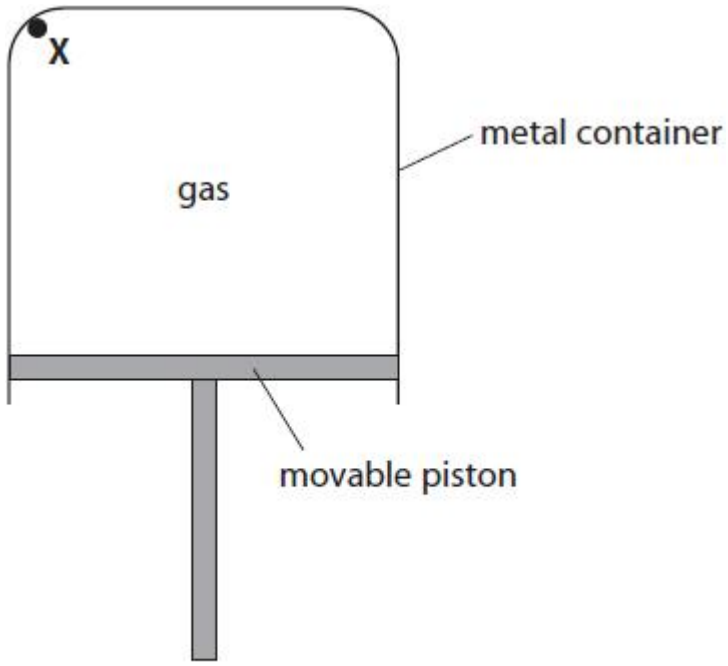

Figure 16

Point X is on the inner surface of the container.

The gas in the container is at a higher pressure than the air outside the container.

The pressure of the gas in Figure 16 (P_1) is 120 kPa.

The volume of the gas in Figure 16 (V_1) is 2500 cm^3.

The piston is pushed up slowly so that the temperature of the gas does not change.

The new volume of the gas (V_2) is 1600 cm^3.

Calculate the new pressure of the gas, P_2.

Use the equation

$$P_2 = \frac{P_1 \times V_1}{V_2}$$

(2)

new pressure, P_2 = ... kPa

(Total for question = 2 marks)

Q5.

Figure 21 shows some water in a measuring cylinder and a lump of iron.

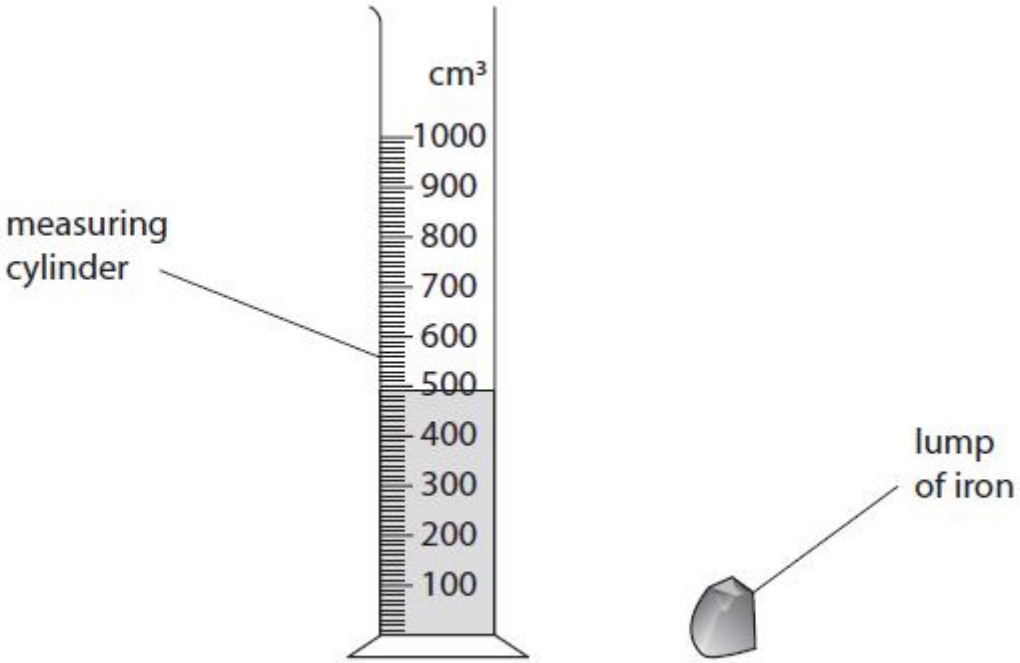

Figure 21

The lump of iron is lowered fully into the water.

The water level in the measuring cylinder rises to 530 cm^3.

The density of iron is 7.9 g / cm^3.

Calculate the mass of the lump of iron.

Use the equation

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

Give your answer to 2 significant figures.

(4)

mass = ... g

(Total for question = 4 marks)

Q6.

Figure 14 shows a saucepan of milk being heated on an electric cooker.

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Figure 14

Figure 15 is a table of data about the milk being heated.

mass of milk	0.82 kg
starting temperature of milk	10 °C
final temperature of milk	40 °C
change in thermal energy of milk	96 000 J

Figure 15

(i) Using data from the table in Figure 15, calculate the increase in temperature of the milk.

(1)

increase in temperature = ... °C

(ii) Using data from the table in Figure 15, calculate the specific heat capacity of the milk.

Use the equation

$$\text{specific heat capacity} = \frac{\text{change in thermal energy}}{\text{mass} \times \text{increase in temperature}}$$

(2)

specific heat capacity = ... J/kg °C

(Total for question = 3 marks)

Q7.

Figure 20 shows the dimensions of a solid block of concrete.

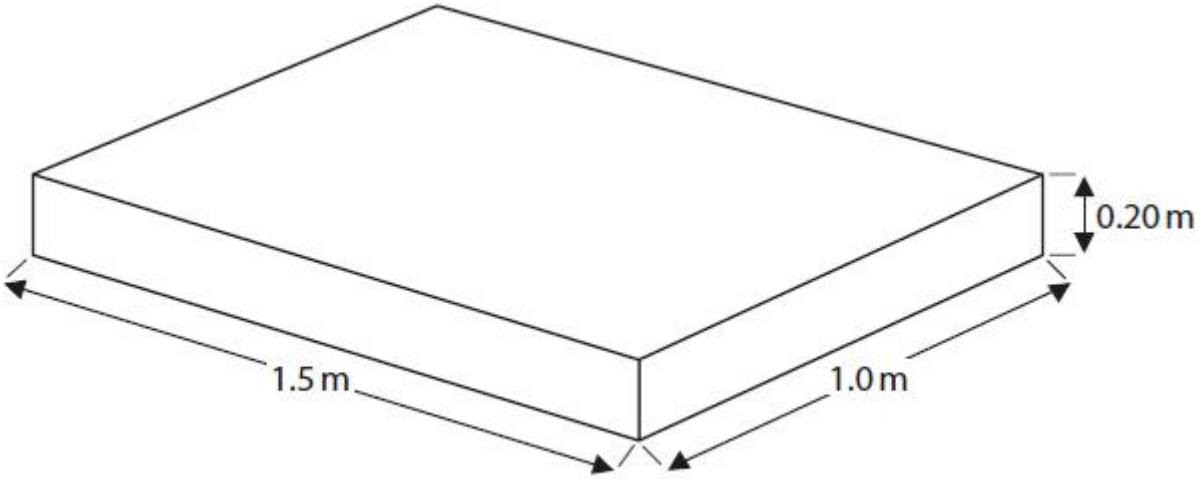

Figure 20

Density of concrete, ρ, = 2100 kg / m^3.

Calculate the mass of the concrete block.

Use the equation:

$$m = \rho \times V$$

(3)

mass of concrete block = ... kg

(Total for question = 3 marks)

Q8.

(i) Figure 11 shows an electric kettle.

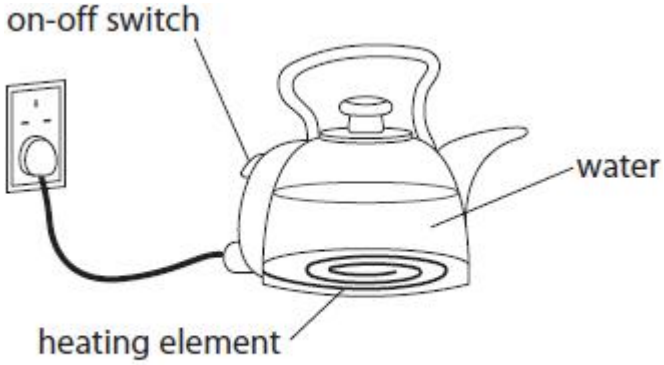

Figure 11

The kettle contains 1.5 kg of water.
The kettle is switched on.
Calculate the energy needed to raise the temperature of the water by 50 °C.
Specific heat capacity of water = 4200 J/kg °C
Use the equation

$$\Delta Q = m \times c \times \Delta\theta$$

(2)

energy needed = ... J

(ii) The amount of energy, *E*, needed to bring the water to boiling point is 670 000 J.

The kettle has a power of 3500 W.
Calculate the time, *t*, it takes to bring the water to boiling point.
Use the equation

$$P = \frac{E}{t}$$

(3)

time to bring the water to boiling point = ... s

(Total for question = 5 marks)

Q9.

An object has a mass of 7.22×10^{-2} kg and a volume of 2.69×10^{-5} m^3.

Calculate the density, ρ, of the object.

Use the equation

$$\rho = \frac{m}{V}$$

(3)

State the unit.

density =..... unit .....

(Total for question = 3 marks)

Q10.

Figure 19 shows a small piece of copper about 3 cm high.

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Figure 19

A student wants to determine the density of copper.

The student uses a balance to measure the mass of the piece of copper.

(i) Explain how the student could measure the volume of the piece of copper.

(3)

.....

.....

.....

.....

.....

.....

(ii) The mass of the piece of copper is 0.058 kg.

The volume of the piece of copper is 6.5×10^{-6} m^3.
Calculate the density of copper.

(2)

density of copper = ... kg/m^3

(Total for question = 5 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
	substitution (1) 566 000 = 0.25 x L rearrangement (1) $\frac{566\ 000}{0.25}$ evaluation (1) 2 260 000 (J/kg)	substitution and rearrangement in either order award full marks for correct answer without working	(3)

Q2.

Question Number	Answer	Additional guidance	Mark
	substitute (1) 8.00 x 14.5 = P_2 x 1160 rearrangement (1) $\frac{8.00 \times 14.5}{1160}$ (=P_2) evaluation 0.1 (MPa)	Allow 8.00 x 14.5 =116 for one mark award full marks for the correct answer without working	(3)

Q3.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) $P_1 = \frac{105 \times 2.3}{0.2}$ evaluation (1) P_1 = 1200 (kPa)	 allow values that round to 1200 e.g. 1207.5 award full marks for the correct answer without working	**(2)** **AO2**

Question number	Answer	Additional guidance	Mark
(ii)	Use relevant information from table (1) relevant calculation (1) **either** (volume of 30 balloons =) 0.07 x 30 **or** (number of balloons =) $\frac{2.3}{0.07}$ **or** (volume per balloon=) $\frac{2.3}{30}$ comparison / supported conclusion (1) 2.1 is less than 2.3 Or 32 is more than 30 Or 0.077 is more than 0.07	2.3 used in a calculation or comparison 2.1 (m^3) scores MP2 only 32(.8) scores MP1 and MP2 0.077 (m^3) scores MP1 and MP2 32 therefore claim is correct	**(3)** **AO3**

Q4.

Question number	Answer	Additional guidance	Mark
	substitution (1) $(P_2 =)\ \frac{120 \times 2500}{1600}$ evaluation (1) 190 (kPa)	 award full marks for the correct answer without working accept values that round to 190; e.g.187.5, 188, 187	**(2)**

Q5.

Question number	Answer	Additional guidance	Mark
	calculation of change in volume (1) (530 cm^3 – 490 cm^3) = 40 (cm^3) substitution (1) $7.9 = \frac{mass}{40}$ rearrangement and evaluation (1) (mass = 7.9 x 40) (mass =) 316 (g) evaluation to 2 sig fig (1) 320 (g)	measurement mark – using scale allow use of incorrect volume answers without working 316 scores 3 marks 0.316 kg scores 3 marks 316 to any other power of 10 scores 2 marks 4187 or 3871 scores 2 marks (incorrect volume) any answer written to 2sf independent mark answers without working 320 scores 4 marks 320 to any other power of ten scores 3 marks 4200 scores 3 marks 3900 scores 3 marks	**(4)** **A02**

Q6.

Question	Answer	Additional guidance	Mark
(i)	30 (°C) (1)		**(1)** **AO3.1**

Question	Answer	Additional guidance	Mark
(ii)	substitution (1) (c =) $\frac{96\,000}{0.82 \times 30}$ evaluation (1) (c =) 3900 (J/kg °C)	allow ECF from (a)(i) throughout allow values that round to 3900 e.g. 3902.4 (J/kg °C) award full marks for the correct answer without working	**(2)** **AO3.1**

Q7.

Question number	Answer	Additional guidance	Mark
	volume substitution (1) 1.5 x 1.0 x 0.2(0) (= 0.3)		**(3)** **AO2**
	substitution in equation (1) mass = 2100 x (0.3(0))	ecf from calculated value of volume for this mark only	
	evaluation (1) = 630 (kg)	award 2 marks for 6.3 x any other power of 10 5670 gains 1 mark from use of 1.5+1.0+0.2=2.7 award full marks for correct answer without working	

Q8.

Question Number	Answer	Additional guidance	Mark
(i)	substitution (1) $(\Delta Q) = 1.5 \times 4200 \times 50$ evaluation (1) 320 000 (J)	 accept 315 000 (J) 310 000 (J) award full marks for the correct answer without working 320 000 000 315 000 000 310 000 000 score 1 mark (mass in grams)	(2)

Question Number	Answer	Additional guidance	Mark
(ii)	substitution (1) $3500 = \frac{670\,000}{t}$ rearrangement (1) $(t=) \frac{670\,000}{3500}$ evaluation (1) 190(s)	accept substitution and rearrangement in either order accept any answer that round to 190(s) power of ten error award 2 marks maximum award full marks for the correct answer without working	(3)

Q9.

Question number	Answer	Additional guidance	Mark
	substitution (1) $(r) = \frac{7.22(\times 10^{-2})}{2.69(\times 10^{-5})}$ evaluation (1) (ρ =) 2680 unit (1) kg / m^3	 2.68 to any power of ten seen allow any value that rounds to 2680; e.g. 2684 accept 2700 allow values in standard form e.g. 2.68 x 10^3 kg m^{-3}	**(3)** **AO2**

		allow for three marks: 2.68 to any power of ten **with** a consistent unit, e.g. 2680 kg/m^3 2680 g/dm^3 2.68 g/cm^3 2.68 kg/dm^3 0.00268 kg/cm^3 2 680 000 g/m^3 allow for two marks: • 2680 with no or incorrect unit • 2.68 to any other power of 10 **with** an inconsistent unit of density • correct substitution **with** an inconsistent unit of density allow for one mark: • 2680 to any other power of ten with no or incorrect unit • appropriate unit of density with no or an incorrect value	

Q10.

Question number	Answer	Additional guidance	Mark
(i)	an explanation linking any three of the following : use a measuring cylinder /beaker or use a eureka can /displacement can/container with spout (1) (partly) fill measuring cylinder /beaker (with water) note the reading or fill (eureka) can to spout (1) immerse piece of copper (in water) (1) note difference in readings of water level (in measuring cylinder /beaker) or collect water from spout in a measuring cylinder /beaker (1)	give credit for other acceptable methods If no other marks scored then allow 1 mark for attempt to measure volume directly: e.g. fill copper tube with water, tip out and measure volume or measure dimension(s) of copper tube	(3)

Question number	Answer	Additional guidance	Mark
(ii)	recall and substitution (1) density= $\frac{m}{V}$ (density=) $\frac{0.058}{6.5\ (\times 10^{-6})}$ evaluation (1) 8.9×10^{3} (kg/m^{3})	 accept values that round to 8900 e.g. 8923(kg/m^{3}) or 9000 8.9 to any other power of ten gains 1 mark award full marks for correct answer without working.	**(2)**